CONFIG REFERENCE

CONFIGURATION REFERENCE

Copy config/cluster.conf.example to cluster.conf.

k3s platform notes

- K8S_DISTRO="k3s" for remote lightweight path.
- Include --disable-kube-proxy in K3S_EXEC_ARGS with Cilium.
- Install order: **Cilium ? cert-manager ? MetalLB ? KubeVirt/CDI ? metrics-server**.
- Healthy re-deploy skips k3s reinstall (FORCE_K3S_REINSTALL=1 to replace).
- Tested: KubeVirt **v1.8.x** on k8s **v1.35.x** (k3s).

Cluster identity

Variable · Default · Notes
CLUSTER_NAME · mycluster

Kubespray source

Variable · Default · Notes
KUBESPRAY_VERSION · v2.31.0
KUBESPRAY_DIR · .kubespray · relative to this repo root

Kubernetes distribution

Variable · Default · Notes
K8S_DISTRO · kubespray · kubespray \ · k3s
K3S_DATA_DIR · /data/k3s
K3S_VERSION · _empty_ · e.g. v1.35.5+k3s1 (empty = latest channel)
K3S_EXEC_ARGS · --flannel-backend=none --disable-network-policy --disable traefik --disable servicelb --disable-kube-proxy
CILIUM_HUBBLE_ENABLED · true
CILIUM_KUBE_PROXY_REPLACEMENT · true
CILIUM_BGP_ENABLED · false
CILIUM_WIREGUARD_ENABLED · false
CILIUM_GATEWAY_API_ENABLED · false
CILIUM_K8S_SERVICE_HOST · 127.0.0.1 · multi-node: set to first control-plane IP (not 127.0.0.1)
PLATFORM_STACK_ENABLED · true
CERT_MANAGER_ENABLED · true
METALLB_ENABLED · true
METALLB_IP_POOL · _empty_ · required when METALLB_ENABLED=true, e.g. "192.168.1.240-192.168.1.250"
CDI_ENABLED · true
KUBEVIRT_CSI_ENABLED · true
KUBEVIRT_DISKS_PROVIDER · true
SNAPSHOT_CONTROLLER_ENABLED · true
CNAO_ENABLED · true

SSP_ENABLED · true
LONGHORN_ENABLED · false
DESCHEDULER_ENABLED · false
NFD_ENABLED · false
GPU_OPERATOR_ENABLED · false

Kubernetes (Kubespray path)

Variable · Default · Notes
KUBE_VERSION · v1.35.4
CONTAINER_MANAGER · podman · podman \ · containerd \ · docker
CNI_PLUGIN · cilium · cilium \ · calico \ · flannel \ · kube-ovn

KUBE_PODS_SUBNET · 10.233.64.0/18

KUBE_SERVICE_ADDRESSES · 10.233.0.0/18

Node SSH access

Variable · Default · Notes
SSH_USER · root
SSH_KEY · $HOME/.ssh/id_rsa
SSH_PORT · 22 · set to e.g. 2222 when nodes are containers with port forwarding
SSH_EXTRA_ARGS · _empty_ · e.g. "-o StrictHostKeyChecking=no"

Nodes (space-separated IPv4 addresses)

Variable · Default · Notes

CONTROL_PLANE_IPS · 192.168.1.100

WORKER_IPS · 192.168.1.101 192.168.1.102

Optional add-ons

Variable · Default · Notes
METRICS_SERVER_ENABLED · true
INGRESS_NGINX_ENABLED · false
DASHBOARD_ENABLED · false

KubeVirt ? VM workloads on Kubernetes

Variable · Default · Notes
KUBEVIRT_ENABLED · true
KUBEVIRT_VERSION · v1.3.1 · k3s platform: v1.8.x tested on k8s v1.35.x
CDI_VERSION · v1.59.0 · k3s platform: v1.65.x tested with KubeVirt 1.8

kubeconfig output

Variable · Default · Notes
KUBECONFIG_OUTPUT · $HOME/.kube/${CLUSTER_NAME}.yaml
KUBECONFIG_MERGE_DEFAULT · false · merge into ~/.kube/config after deploy (k3s)

Secondary disk for KubeVirt VM PVCs (keeps root/k3s data on primary volume).

Variable · Default · Notes
LAB_SECONDARY_DISK · _empty_ · e.g. /dev/sdb ? empty skips lab storage
LAB_KUBEVIRT_VOLUME_MOUNT · /data/kubevirt-volumes
LAB_KUBEVIRT_VOLUME_SUBPATH · /data/kubevirt-volumes/pvc
LAB_KUBEVIRT_STORAGE_CLASS · kubevirt-sdb
LAB_STORAGE_FORMAT · 0 · 1 = mkfs empty LAB_SECONDARY_DISK (destructive)
LAB_MEMORY_TUNE_ENABLED · false
LAB_AUTO_SETUP · false